Supplementary Material

Causal Transfer Learning: Warm-Starting NOTEARS, DAGMA, and GOLEM with Cross-Dataset Graphs

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This document contains supporting analyses, full implementation details, and per-configuration results that accompany the main text. Section S3 provides the edge-stratification table requested in review; Section S5 reports per-seed genome-scale results. References to sections (S1–S12) and tables (S1–S14) refer to this document unless stated otherwise.

S1. Biological universality of causal programs

The transfer results rest on a falsifiable hypothesis: causal gene regulatory programs contain a conserved backbone that recurs across tissue contexts. The hypothesis draws on three lines of prior evidence.

First, the TCGA Pan-Cancer Atlas established that molecular subtypes—and the transcriptional programs that define them—transcend tissue-of-origin boundaries.[1] If expression-based molecular subtypes are shared, the regulatory programs that generate those expression patterns may also be shared.

Second, essentiality screens[2] identify a core set of ∼1,500 fitness genes whose knockout impairs viability across hundreds of cell lines. The regulatory logic governing these essential genes is under strong evolutionary constraint and is unlikely to vary across contexts.

Third, developmental signaling pathways (Wnt, Notch, Hedgehog, TGF-β) are co-opted in cancer with conserved downstream transcriptional targets.[3] The transferred edges—including WNT2→ C2orf40—belong to precisely these pathways.

Taken together, these observations predict that a subset of regulatory edges should recur across cancer types, and that this subset should be enriched for core programs rather than for tissue-specific ones. The warm-start advantage reported in the main text is consistent with this prediction: the transferred edges are concentrated in conserved programs, whereas the edges that fail to transfer are predominantly dataset-specific or weakly supported.

S2. High-dimensional extension ($d = 5{,}000$): implementation details

The main text reports that warm-starting preserves $81.8 \pm 0.4\%$ of source edges at $d = 5{,}000$, versus $45.4 \pm 2.9\%$ from scratch across three seeds, both evaluated against each seed's own source graph. This section documents the computational and numerical choices behind that result.

Optimizer. At $d = 5{,}000$ the parameter matrix has 25 million entries whose per-parameter curvature history would be memory-prohibitive for L-BFGS-B, so the inner optimizer is ADAM (lr = 0.002, full-batch gradients) rather than the L-BFGS-B used at $d \leq 889$. The augmented-Lagrangian outer loop uses 20 iterations with $\rho_0 = 1$, a larger initial penalty than the $\rho_0 = 0.1$ of the main experiments, chosen to accelerate DAG enforcement at this scale. All other hyperparameters ($\lambda = 0.01$, edge threshold $\tau = 0.3$) match the main experiments.

Numerical stability. The acyclicity term $\mathrm{tr}(\exp(W \circ W))$ is evaluated in float32, where the matrix exponential overflows once the row-sum of $W \circ W$ approaches $\sim$30. Source-graph weights are therefore bounded so that this row-sum stays below that bound (here 1.39), and scratch training uses random initialization $\mathcal{N}(0, 0.05^2)$ across seeds (42, 123, 456), reduced from the 0.1 used at $d \leq 889$ because 0.1 would place the row-sum near 50. Each seed learned its own source graph from that random initialization, giving 624 ± 4 source edges at $\tau = 0.3$ (range 619–627); the warm-start run for a seed uses that seed's own source graph, as in the $d \leq 889$ benchmark. For reference, a single deterministic run initialized at $W = 0$ yields a denser 634-edge source graph and a warm retention of $513/634 = 80.9\%$, consistent with the seed-matched estimate.

Compute. NOTEARS converged on a single consumer GPU (NVIDIA RTX 5060 Laptop, 8 GB) using 7.7 GB of VRAM and 3.5 hours per run, with the $\mathcal{O}(d^3)$ matrix exponential as the sole computational bottleneck. Warm-starting also recovers more total edges than scratch (842 ± 10 vs. 703 ± 9).

Relation to the scaling law. Retention is necessarily bounded by 100%, so the power law $r(d) \propto d^{0.47}$ established at $d \leq 889$ cannot be extrapolated indefinitely; the measured $d = 5{,}000$ warm retention ($81.8 \pm 0.4\%$) is accordingly below a naive extrapolation, while the absolute advantage over scratch (+36.4 percentage points, versus +34.0 pp at $d = 889$, the value in Table 2) remains substantial and consistent with the trend of growing transfer benefit with dimension.

Extrapolation beyond $d = 5{,}000$. Extrapolating the cubic time scaling from the measured run gives $d = 10{,}000$ at $\sim$28 hours ($8\times$ the $d = 5{,}000$ runtime) and $d = 16{,}000$ at roughly 5 days. Assuming quadratic memory scaling—a conservative upper bound—$d = 10{,}000$ would require $\sim$31 GB and $d = 16{,}000$ $\sim$79 GB, beyond an 8-GB consumer GPU but within reach of 24-GB or multi-GPU systems. Low-rank NOTEARS with rank $r \approx 100$ reduces the bottleneck to $\mathcal{O}(d\,r^2)$, bringing genome-scale causal transfer within reach of a single GPU.

Why the log-determinant term is rigid. The main text compresses the DAGMA mechanism to three sentences; the full argument is as follows. DAGMA's characteristic loss term $-\log|\det(I - W)|$ contributes the gradient $\nabla_W[-\log|\det(I - W)|] = (I - W)^{-\top}$, which is bounded at a feasible source solution: for a DAG the matrix $I - W_S$ is unit-triangular in topological order, so every eigenvalue equals one and $\det(I - W_S) = 1$. Any perturbation that would close a cycle drives $\det(I - W) \to 0$, where the inverse—and hence the gradient—diverges. The log-det term therefore acts as a one-sided barrier: remaining at W_S is cheap, leaving it is expensive, which is precisely the local rigidity that yields dimension-independent retention. (Acyclicity itself is enforced by the trace-exponential constraint; the log-det term is the additional penalty that shapes the local geometry.) The MSE loss of NOTEARS has the opposite geometry: its Hessian, $\frac{1}{n}X^\top X$ up to the DAG penalty, has a spectrum spanning orders of magnitude on transcriptomic covariates, so the objective is nearly flat along many directions and admits a combinatorially growing set of near-equivalent solutions as d increases. Warm-starting thus supplies a bias that the MSE objective cannot overturn, whereas the augmented Lagrangian term $h(W)^2$ exerts only gradual pressure toward the DAG constraint. The power-law exponent 0.47 follows: as d grows, the number of near-equivalent MSE solutions increases combinatorially, so the warm-start bias becomes proportionally more valuable.

Pathway mapping of the seven transferred DepMap edges. The DepMap→BRCA transfer retains seven edges at $\tau = 0.3$ (35% of the 20 source edges; full list in `results/depmap_edges_d454.json`). Their gene annotations map to calcium homeostasis (S100P→PPP1R14C), epithelial differentiation (POF1B→KRT16), cell adhesion and metastasis (SLC44A4→TSPAN8→PCSK9), Notch signaling and antioxidant defence (HES2→GPX2), protein processing (PCSK9→SERPINB5) and ubiquitin-mediated proteolysis (USP9Y→POF1B). The calcium-homeostasis edge S100P→PPP1R14C is preserved under both the TCGA and DepMap transfers, marking a core relationship conserved across measurement platforms.

S3. Edge stratification and pathway membership

Table S3 stratifies the 73 source edges of the TCGA→BRCA transfer at $\tau = 0.3$ by source-graph weight $|W_S|$. For each band it reports the number of edges, the retention achieved by scratch and by warm-start at the same threshold, and the KEGG pathways most represented among the genes touched by the band. Warm-start matches or exceeds scratch in every band. KEGG annotation was resolved through the KEGG REST API (`/find/hsa/` for symbol→entrez and `/link/pathway/` for pathway membership); 51 of the 105 genes touched by source edges carried at least one KEGG pathway annotation.

Table S3: Edge-strength stratification at $\tau = 0.3$ ($d = 454$, TCGA→BRCA; 73 source edges). Retention is the fraction of band edges recovered at $|W| > 0.3$.

Band $\lvert W_S \rvert$	n	Scratch (%)	Warm (%)	Δ (pp)
$[0.3, 0.4)$	25	24.0	44.0	+20.0
$[0.4, 0.5)$	18	16.7	33.3	+16.6
$[0.5, 0.6)$	8	62.5	62.5	0.0
$[0.6, 0.7)$	10	80.0	90.0	+10.0
$[0.7, 1.0)$	12	50.0	83.3	+33.3
All	73	38.4	56.2	+17.8

Predominant KEGG pathways per band (gene counts in parentheses). Full per-gene annotation is stored in `results/edge_stratification_tau03.json`.

- $[0.3, 0.4)$: Metabolic pathways (6), Cornified envelope formation (4), Cadherin signaling (4), Chemical carcinogenesis–reactive oxygen species (3), Protein digestion and absorption (2), Glutathione metabolism (2), Estrogen signaling (2), *Staphylococcus aureus* infection (2).
- $[0.4, 0.5)$: Metabolic pathways (4), Cornified envelope formation (3), Retinol metabolism (2), Metabolism of xenobiotics by cytochrome P450 (2), Drug metabolism–cytochrome P450 (2), Cytokine–cytokine receptor interaction (2), Prostate cancer (2), Cadherin signaling (2).
- $[0.5, 0.6)$: Cornified envelope formation (2), Cadherin signaling (2), Estrogen signaling (2), *Staphylococcus aureus* infection (2), TGF-β signaling (1), Glycosaminoglycan biosynthesis–heparan sulfate (1), Renin–angiotensin system (1), Endocrine calcium reabsorption (1).
- $[0.6, 0.7)$: Cornified envelope formation (3), Cadherin signaling (3), Transcriptional misregulation in cancer (2), MicroRNAs in cancer (2), Estrogen signaling (2), *Staphylococcus aureus* infection (2), Choline metabolism in cancer (2), Cytoskeleton in muscle cells (1).

- $[0.7, 1.0)$: Metabolic pathways (3), Cadherin signaling (3), Mucin type O-glycan biosynthesis (1), Glycosphingolipid biosynthesis (1), Lysine degradation (1), Neuroactive ligand–receptor interaction (1), Neuroactive ligand signaling (1), Retrograde endocannabinoid signaling (1).

Pathway membership is reported as annotation rather than as an enrichment statistic: only two of the 73 source edges touch the canonical *Pathways in cancer* map (hsa05200), so this stratification is not powered for pathway-level claims, and the main text accordingly reports transfer rates by edge strength rather than by pathway.

S4. External validation: database-level detail

Validation was performed on the top-200 edges by absolute weight for each method ($d = 454$). STRING v12 was queried at combined-score thresholds 400, 700 and 900; mapping succeeded for 97.8% of genes. TRRUST, RegNetwork and PathwayCommons are curated resources recording experimentally supported transcription-factor/target and protein–protein interactions, and therefore overlap only marginally with co-expression-inferred associations.

Table S4: External validation detail ($d = 454$, top-200 edges per method). Percentages are the fraction of mappable edges supported by each database.

Method	STRING ≥ 400	STRING ≥ 700	STRING ≥ 900	BioGRID	TRRUST
TCGA source	4.6	3.0	2.0	0.5	0.0
Scratch	4.5	3.0	2.5	1.0	0.0
TCGA-warm	4.5	3.0	2.5	1.0	0.0
DM-warm	4.5	3.0	2.5	1.0	0.0

BioGRID v5.0.260 was queried for the same edge sets, yielding 1.0% precision (2/200 edges) for scratch, TCGA-warm and DepMap-warm, and 0.5% (1/200) for the TCGA source graph. The two BioGRID-confirmed pairs are KRT16–KRT7 and PRSS3–AGR2. RegNetwork and PathwayCommons yield 0% across all methods, as does TRRUST. The five preserved edges central to the cross-domain analysis (WNT2→C2orf40, S100P→PPP1R14C, PRKY→ATP2C2, LGI3→SLC26A9, POF1B→SLC44A4) are absent from BioGRID, consistent with their status as *de novo* discoveries.

The relevant comparison is not the absolute precision but its invariance across methods: scratch and both transfer variants reach essentially the same precision at every threshold. Had warm-starting manufactured spurious edges, precision would have fallen; had it merely re-derived database-annotated edges, precision would have risen. Neither occurs.

S5. Per-seed genome-scale results

Table S5 reports the $d = 5{,}000$ experiment at the level of individual seeds. The protocol matches the $d \leq 889$ benchmark exactly: each seed (i) learns its own source NOTEARS graph on the 9,241 non-breast samples from a random initialization $\mathcal{N}(0, 0.05^2)$, and (ii) runs both a warm-start (initialized at that source graph) and a scratch run (initialized from the same distribution) on the 1,218 BRCA samples. Retention is the fraction of that seed's source edges recovered at $|W| > 0.3$, so numerator and denominator are always seed-matched; the earlier single-run figure of $513/634 = 80.9\%$ is superseded by this protocol. Because the source graph itself differs across

seeds (619–627 edges), the denominator is not fixed, which is the same reason the $d \leq 889$ tables carry non-zero variance in both columns.

Table S5: Per-seed genome-scale transfer (d = 5,000, τ = 0.3). Retention for every method is measured against the same seed's source graph.

Seed	Source edges	Warm edges	Warm kept (ret.)	Scratch edges	Scratch kept (ret.)	Δ (pp)
42	619	856	505 (81.6%)	702	299 (48.3%)	+33.3
123	626	833	510 (81.5%)	693	290 (46.3%)	+35.1
456	627	838	516 (82.3%)	714	260 (41.5%)	+40.8
Mean±SD	624 ± 4	842 ± 10	$81.8 \pm 0.4\%$	703 ± 9	$45.4 \pm 2.9\%$	+36.4

Two observations follow. First, warm retention is remarkably stable across seeds (SD 0.4 pp), whereas scratch retention varies by 6.8 pp: the warm-start solution is anchored by the source structure, while scratch converges to a different near-equivalent DAG on each run. Second, the per-seed gain is positive for every seed and increases as scratch degrades (+33.3, +35.1, +40.8 pp), i.e. the benefit is largest precisely when the unguided optimization lands furthest from the source structure.

Relation to the single-run figures in the first response. The first response reported a single warm-start value, 513/634 = 80.9%, and a three-seed scratch value of $38.1 \pm 1.4\%$. Both were measured against one deterministic source graph obtained from a zero initialization; the warm-start figure consequently had no seed-to-seed variance, and neither figure corresponds to the seed-matched denominator used throughout the $d \leq 889$ tables. Evaluating the present warm runs against that same 634-edge reference instead gives $65.3 \pm 0.7\%$ versus $38.1 \pm 1.4\%$ (+27.2 pp), lower than the seed-matched 81.8% because each seed's own source graph overlaps the reference graph by only 77–79%, and $0.78 \times 0.82 \approx 0.65$. The seed-matched figures in Table S5 are the ones that answer the reviewer's request, since they are the quantities reported for every other dimension; the direction and the significance of the conclusion are unchanged under either denominator, all three seeds gaining 27–41 percentage points. Per-seed matrices and the consolidated statistics are stored in `results/d5000_seedmatched.json`, `results/d5000_multiseed_transfer.json` and `results/d5000_checkpoints/`.

S6. Reproducibility

All experiments use fixed random seeds (42, 123, 456). Data are public: TCGA via the UCSC Xena browser (`https://xenabrowser.net/`), DepMap (`https://depmap.org/`), STRING v12 (`https://string-db.org/`), and TRRUST v2 (`https://www.grnpedia.org/trrust/`). Source code and pre-computed causal graphs are available at `https://github.com/sgao-academics/causal-transfer-discovery`. The main-text analyses are reproduced by `scripts/run_all.py`; the genome-scale extension by `scripts/run_d5000_multiseed.py`; the multi-organ extension of Section S12 by `scripts/experiments/run_multi_target.py` (d = 454) and `scripts/run_d5000_multitarget.py` (d = 5,000).

Table S7: Edge retention at varying thresholds ($d = 454$, TCGA→BRCA). Counts are source-graph edges at that threshold that are also present in the scratch or warm graph at the same threshold.

τ	Source edges	Scratch overlap	Warm overlap	Δ
0.1	1156	122 (11%)	158 (14%)	+36
0.2	168	38 (23%)	57 (34%)	+19
0.3	73	28 (38%)	41 (56%)	+13
0.4	48	20 (42%)	28 (58%)	+8
0.5	30	13 (43%)	20 (67%)	+7

S7. Sensitivity to the edge threshold

Pooling the $\tau \geq 0.5$ bands of Table S3 (the 30 source edges with $|W_S| > 0.5$) reproduces the $\tau = 0.5$ row of Table S7 (warm $20/30 = 67\%$, scratch $13/30 = 43\%$), confirming that the strength-stratified and threshold-stratified views are consistent.

S8. Paired statistics for the core comparison

Every main-text comparison is paired: the warm-start and scratch runs for a given (d, seed) share the same gene set, the same target samples and the same optimizer, so the appropriate test is a paired one. Table S8 gives the per-dimension paired differences for NOTEARS. Across the 15 NOTEARS pairs the mean gain is +26.3 pp (Wilcoxon signed-rank $p = 6.1 \times 10^{-5}$; exact sign test 15/15, $p = 3.1 \times 10^{-5}$), and the smallest single gain is +7.9 pp at $d = 200$.

DAGMA and GOLEM use the same protocol. The two additional algorithms in Table 4.4 of the main text are run identically: each seed $s \in \{42, 123, 456\}$ learns its own source graph on the 9,241 non-breast samples, the warm arm is initialised from that graph, the scratch arm is initialised from $\mathcal{N}(0, 0.1^2)$, and both are scored against the *same* source graph. The denominators are therefore seed-matched, as for NOTEARS; the three seeds differ in the source graph and hence carry genuine variance in both arms. This is what makes the per-dimension standard deviations in Table 4.4 meaningful and what makes the 45 configurations 45 independent comparisons. Across all 45 algorithm–dimension–seed configurations warm-start exceeds scratch (sign test 45/45, $p = 2.8 \times 10^{-14}$). The shuffled control, by contrast, is indistinguishable from scratch (paired Wilcoxon $p = 0.86$, mean difference +0.2 pp), while warm-start exceeds shuffled at all 15 pairs ($p = 6.1 \times 10^{-5}$).

Table S8: Per-dimension paired differences (NOTEARS, $\tau = 0.3$, 3 seeds). The difference column is warm minus scratch within the same (d, seed).

d	Scratch (%)	Warm (%)	Paired Δ (pp)	min	max	wins
200	19.9	31.9	+12.0	+7.9	+17.2	3/3
454	33.4	55.1	+21.7	+17.8	+24.4	3/3
600	33.3	66.6	+33.3	+27.6	+39.3	3/3
750	36.5	66.8	+30.3	+25.7	+35.6	3/3
889	34.0	68.1	+34.0	+29.9	+36.5	3/3
All	—	—	**+26.3**	+7.9	+39.3	**15/15**

Standard-deviation convention. All standard deviations in the main text and in this document are sample standard deviations $(n-1)$. Reporting the population convention (n) would reduce every value by the factor $\sqrt{(n-1)/n} = 0.816$ (18% at $n = 3$); we state the convention explicitly, and every mean, SD and paired difference can be recomputed from `results/benchmark_v6_checkpoint.json`.

Post-hoc thresholding. The acyclicity constraint is enforced on the unthresholded matrix, where $h(W) < 10^{-6}$ by construction. Thresholding at $\tau = 0.3$ is applied only to obtain an interpretable edge set for edge counts, retention and database validation, and (as is generally the case for post-hoc thresholding of continuous DAG estimators) the thresholded graph need not itself be acyclic.

S9. Is pooling a substitute for transfer?

If the source samples are available, a referee will ask why the cohorts are not simply concatenated and one graph fitted to the union. We ran that arm: the 9,241 source and 1,218 BRCA samples are stacked, standardised jointly, and a single NOTEARS graph is estimated with the same penalty, threshold and outer-loop budget as every other arm, then scored against the same source graph.

Retention is not comparable across arms here. Pooling sees 88% of the samples that generated the source graph, so its retention is high by construction; in the other direction, $\tau = 0.3$ is an absolute cut on weights whose scale shrinks as n grows, so the number of edges surviving the cut falls as samples are added. Neither effect reflects structural agreement. The comparison that survives both problems is the held-out residual of the thresholded graph on the 20% of BRCA withheld from every arm, computed after every graph has been cut to the *same* number of edges k: at fixed k the residual measures how well each edge set explains breast-tissue covariance, and the complexity penalty is identical by construction.

Table S9: Pooling the source and target samples is not a substitute for transfer. One graph is fitted to the stacked matrix of all 9,241 source and 1,218 BRCA samples and scored against the same source graph. Retention is not comparable across arms here, because $\tau = 0.3$ is an absolute cut on weights whose scale depends on the sample size; the fair comparison is the held-out residual at a matched edge count k (last two columns, lower is better). Seed 42 shown; all dimensions share that seed.

d	Source edges	Warm ret. (%)	Pool ret. (%)	Warm NRMSE	Pool NRMSE
200	64	29.7	54.7	0.8578	0.8850
454	74	56.8	56.8	0.9251	0.9328
600	85	64.7	51.8	0.9386	0.9419
750	101	69.3	52.5	0.9498	0.9533
889	107	67.3	56.1	0.9540	0.9579

Across all 11 algorithm–dimension–seed cells the pooled graph has a higher held-out residual than the warm graph by +0.012 on average (range +0.003 to +0.031); the penalty is positive in 11 of 11 cells.

Target-only priors do not reproduce the gain. Three further arms build a starting matrix without the source cohort: the |corr| skeleton of BRCA itself, its ridge coefficients, and a random DAG matching the source graph's edge count and out-degree profile. None approaches warm-start's retention, and the degree-matched control tracks scratch almost exactly—so the benefit cannot be attributed to "any sparse hint".

Table S10: Target-only structure priors do not reproduce the transfer benefit. All arms start NOTEARS from a matrix built without the source cohort and are scored against the source graph; means over three seeds.

d	Scratch	Warm	Shuffled	Pool	Corr-prior	Ridge-prior	Degree-matched	k
200	19.9	31.9	17.9	56.0	17.3	16.8	21.5	66
454	33.4	55.1	35.6	55.5	33.8	33.7	36.0	94

At $k = 70$ the six source-free or sparse priors differ from one another by at most 0.007 in held-out residual within a cell (median spread 0.004), whereas the pooled graph sits +0.008 above their mean (median). Predictive residual therefore does not discriminate among sparse priors, but it does penalise the pooled one.

S10. Dependence on the size of the target cohort

The motivating claim of the paper is that transfer matters most when the target cohort is small. Table S11 tests it by subsampling BRCA to $n = 150$, 300, 600 and the full 1,218 samples, holding the source graph, the optimiser, the penalty and the threshold fixed. Warm-start retention is almost flat across a factor of eight in n (55–58%), whereas scratch rises from 21% to 34%: the advantage is +37.2 pp at $n = 150$ against +21.2 pp at $n = 1{,}218$, and it is positive for every seed at every n. What transfer supplies is therefore not extra samples but a starting point whose quality does not depend on how many target samples happen to be available.

Table S11: Retention versus target sample size ($d = 454$, TCGA→BRCA). Each n draws a random subset of the 1,218 BRCA samples; the source graph, optimiser and threshold are unchanged. Mean ± sample SD over three seeds.

n	Scratch (%)	Warm (%)	Δ (pp)
150	21.0±3.4	58.2±4.8	+37.2
300	33.0±6.0	56.0±1.7	+23.0
600	32.1±2.3	57.8±3.9	+25.8
1218	33.9±5.1	55.1±1.7	+21.2

S11. Ten-seed robustness

Table 1 rests on three seeds. Table S12 repeats the NOTEARS comparison on those three plus seven fresh seeds whose random starts are seeded explicitly, leaving the three main-table seeds untouched so that no number in the main text changes. The sign of the warm–scratch difference is unanimous in every configuration reported, and the magnitudes (+6.8 to +29.2 pp at $d = 454$) bracket the three-seed means of the main text.

Table S12: Ten-seed NOTEARS comparison. The first three seeds are those of the main tables; the remaining seven are fresh draws whose source graphs are seeded explicitly. Each row is one seed's own source graph.

	$d = 454$			$d = 889$		
Seed	Scratch	Warm	Δ	Scratch	Warm	Δ
42	32.4	56.8	+24.4	30.8	67.3	+36.5
123	35.6	53.4	+17.8	33.9	69.6	+35.7
456	32.1	55.1	+23.0	37.4	67.3	+29.9
789	40.3	56.9	+16.6	48.1	64.8	+16.7
1011	46.6	53.4	+6.8	47.3	67.3	+20.0
2024	40.0	54.7	+14.7	51.8	67.0	+15.2
3141	48.6	58.3	+9.7	44.5	71.8	+27.3
4242	31.9	61.1	+29.2	47.7	71.0	+23.3
5555	48.5	61.8	+13.3	46.7	69.2	+22.5
6789	38.2	56.6	+18.4	45.6	66.7	+21.1

At $d = 454$ warm-start exceeds scratch in 10 of 10 seeds (exact sign test $p = 2^{-10} = 9.8 \times 10^{-4}$); at $d = 889$, in 10 of 10. Retention rises with dimension for both arms, reproducing the scaling behaviour of the main text on independent draws.

S12. Dependence on the target organ

Every transfer result in the main text uses BRCA as the target, because it is the largest cohort and the one the benchmark was built around. That leaves open the possibility that the advantage is a property of breast cancer rather than of causal transfer. This section repeats the core comparison with five further TCGA cohorts in the target role.

Design. The protocol is leave-one-organ-out. For target organ O the source graph is learned from every other cohort in the panel—BRCA included when $O \neq$ BRCA—so no sample of O enters the prior, and the zero-overlap requirement of Section 3.3 is met by every target rather than only by BRCA. Reusing the archived non-BRCA source graph for all five further targets would have saved roughly 2.5 h per organ, but it would have placed the target's own samples in its prior; that shortcut was not taken. The gene panel is held fixed across organs (the same 454 genes, in the same column order, as the main benchmark), because a per-organ panel would make every column index denote a different gene and destroy comparability of retention. Normalisation, optimiser settings, edge threshold ($\tau = 0.3$) and seeds follow Table 2 exactly: cohort matrices are stacked and z-scored as one matrix, and the target is z-scored on its own.

Retention rises over the scratch baseline in all 18 organ–seed cells (Table S13); Δ spans +16.7 to +30.2 pp with a mean of +22.5 pp (exact sign test $p = 2^{-18} = 3.8 \times 10^{-6}$). The BRCA gain measured here, +23.2 pp, agrees with the +21.7 pp of Table 2.

Table S13: Retention at $d = 454$ with each cohort in the target role. Each target receives its own source graph, learned from all remaining cohorts (leave-one-organ-out). Mean ± sample SD over three seeds; Δ is warm minus scratch, taken per seed and then averaged, so the range column shows the three individual gains rather than the spread of the means.

Organ	n	Scratch (%)	Warm (%)	Mean Δ (pp)	Δ range (pp)
BRCA	1218	37.2 ± 6.2	60.4 ± 4.9	+23.2	+21.9 to +24.7
KIRC	606	40.5 ± 1.0	57.7 ± 1.7	+17.2	+16.7 to +18.1
LUAD	576	43.6 ± 2.0	64.6 ± 3.8	+21.0	+18.0 to +23.3
THCA	572	35.0 ± 3.3	60.3 ± 1.7	+25.3	+24.1 to +27.2
LUSC	553	43.2 ± 3.0	70.9 ± 1.5	+27.8	+25.7 to +30.2
LIHC	423	35.4 ± 4.1	55.8 ± 1.5	+20.3	+17.9 to +24.0

Why the BRCA row does not coincide with Table 2. The scratch and warm columns above sit 4–5 pp away from the Table 2 values (33.4 and 55.1), which needs an explanation rather than a footnote. Re-running the two arms on the *archived* source graphs, instead of re-learning them, reproduces the warm column exactly: 56.8, 53.4 and 55.1% for seeds 42, 123 and 456, mean 55.1%, identical to Table 2 to the tenth of a percentage point. The scratch arm lands one to three edges away (31.1, 39.7, 30.8%), because it starts from a random initialisation drawn at a different position in the RNG stream. The implementation is unchanged; only the draw that fixes the source graph differs.

The consequence is a measurement rather than a caveat. At $d = 454$ the retention *level* carries a source-graph-draw component of roughly ±5 pp, whereas Δ is stable to within about 3 pp; the same component is visible in Table S12, where the seven later seeds scatter appreciably more than the first three. At $d = 5{,}000$ it effectively vanishes (81.8 ± 0.4% over three seeds), because at that dimension the source graph is data-dominated rather than initialisation-dominated. We therefore treat Δ as the organ-comparison quantity at $d = 454$ and do not rank organs by level or by gain: an ordering of six organs separated by 1–10 pp is not resolvable at this noise level.

Five further organs at $d = 5{,}000$. The genome-adjacent experiment of Section 5.4 was repeated with five of these cohorts in the target role, one seed each and the same leave-one-organ-out source construction.

Table S14: Retention at $d = 5{,}000$ with five further cohorts in the target role (seed 42). The BRCA row is the archived three-seed result of Section 5.4 and is shown for reference; the five new rows are single-seed exploratory runs. Every arm reached $h = 0$ on its own between outer iterations 9 and 12.

Organ	n	Source edges	Scratch (%)	Warm (%)	Δ (pp)
BRCA	1218	624	45.4 ± 2.9	81.8 ± 0.4	+36.4
KIRC	606	644	28.6	61.0	+32.5
LUAD	576	632	34.8	77.7	+42.9
THCA	572	626	19.6	57.7	+38.0
LUSC	553	640	29.8	78.3	+48.4
LIHC	423	639	21.3	59.5	+38.2

All five are positive, Δ between +32.5 and +48.4 pp, and no run was truncated: every arm reached $h = 0$ between outer iterations 9 and 12, so the 20-iteration cap of Section S2 was never binding in

any configuration reported here. The spread across organs (57.7 to 78.3% warm retention) lies far outside the 0.4 pp seed noise measured on BRCA, which suggests it reflects the tissue rather than the draw; a single seed per organ cannot establish that, and these rows are therefore reported as an exploratory extension rather than pooled with the three-seed BRCA row.

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